ROHAN BENNUR

Riverside, CA

+1 (951) 425-7354 rbenn020@ucr.edu linkedin.com/in/rohan-bennur-14653210a github.com/rohanbennur43

Education

- **University of California, Riverside** Riverside, California
  Master of Science, Computer Science *Sept. 2024 - Mar. 2026 (expected)*
- **PES University** Bangalore, India
  Bachelor of Technology, Computer Science and Engineering (Machine Intelligence and Data Science) *Aug. 2018 - May 2022*
  - **GPA**: 3.75/4.0
  - **Achievements**: **MRD Scholarship** for being in the top 20 percentile of the class.

Achievements

- **UCR Programming Challenge (October 2024)**: Secured **second place** out of 400 participants in a coding competition that involved tackling complex problems, demonstrating strong skills in data structures and algorithms under time pressure.
- **Open Source Contribution: Librist Project**: Resolved a Content-Type anomaly by conducting an in-depth root cause analysis to address a bug related to inaccurate OpenMetrics assignment during Prometheus metrics exposure. Implemented code enhancements to default the Content-Type to text/plain and dynamically adapted the code to change the Content-Type to OpenMetrics for Prometheus server versions above v2.26.0.

Experience

- **Amagi Media Labs Pvt Ltd**
  Software Engineer *Jan 2022 - July 2024*
  - **Project 1: Scalable Video Processing System**
    * Engineered a **scalable video processing application** using **Golang, Python, and GStreamer** for high-efficiency video data processing.
    * Integrated **Amazon S3** for secure media storage and **MongoDB** for metadata management, improving query performance by **40%** with secondary timestamp indexes.
    * Deployed video processing jobs on **Kubernetes** using **Docker**, PVCs, and ClusterIP services.
    * Reduced ticket turnaround by **25%** with checkpoint-based status updates and optimized RAM and storage by **30%** through workload characterization.
  - **Project 2: Low-Latency Video Streaming Platform**
    * Designed a **low-latency streaming solution** using **RIST protocol and librist**, minimizing buffering and latency.
    * Utilized **gRPC** for interservice communication and dynamic stream switching, and **PubNub queues** for real-time stream quality updates, ensuring seamless delivery and enhanced monitoring.
    * Enhanced system uptime and reduced debugging time by **35%** with **Prometheus and Grafana**.
- **Krushagramati Analytics**
  Research Intern *Oct 2019 - Feb 2020*
  - Researched and developed a **speech emotion recognition system** using **LSTM, CNN, Capsule Networks, and TensorFlow**, addressing hierarchical feature handling and enhancing model generalization.
  - Designed and implemented the **backend with Python Flask** and built a **user-friendly front-end interface with React.js**, ensuring a cohesive user experience.

Projects

- **Ticketing App(Microservices-Based Ticket Purchasing System)**: Devised a ticket purchasing application utilizing a Node.js backend and a React.js frontend, delivering an intuitive user experience. Designed a microservice architecture capable of handling high traffic, employing Docker for containerization, and deploying on Kubernetes for scalability. GraphQL was integrated for efficient API interactions and Redis was utilized for low-latency caching, significantly improving the application performance. Ensured smooth inter-service communication with NATS, facilitating efficient data flow throughout the system while simulating production-level code. Tech: Node.js, React.js, Docker, Kubernetes, GraphQL, Redis, NATS.
- **Clustered Database Indexing Framework (Algorithms, Database Management)**: This project implements a clustered indexing system using a B-Tree data structure to optimize data retrieval and storage efficiency in relational databases. By employing a modular file handling approach with the POSIX API, it effectively manages disk I/O operations and minimizes memory usage. Tech: C, B-Tree, POSIX API, Relational Databases

Certifications

- **TensorFlow Developer Certificate**: Demonstrating proficiency in machine learning and deep learning using TensorFlow, highlighting the ability to design, build, and train advanced machine learning models.